

Producing Halogenoalkanes (Cambridge (CIE) A Level Chemistry): Revision Note

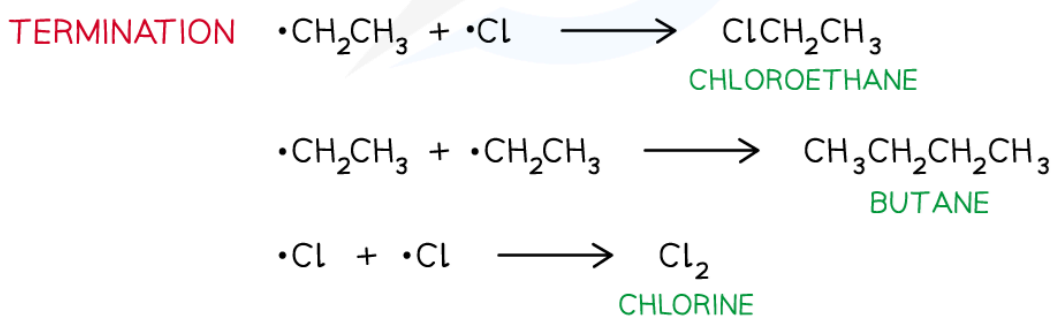
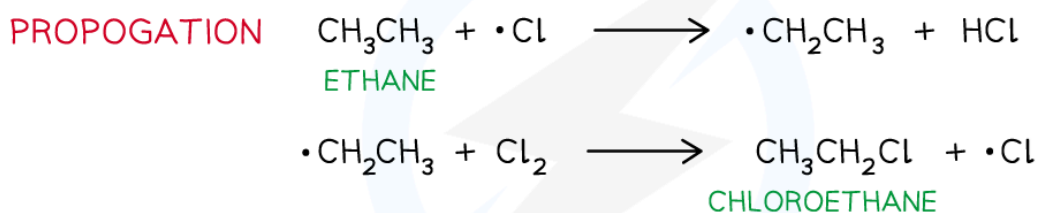
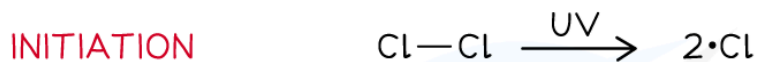
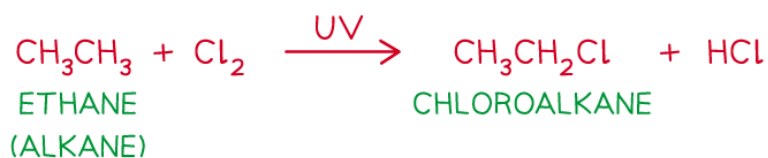
Production of Halogenoalkanes: Substitution & Addition Reactions

- **Halogenoalkanes** are alkanes that have one or more halogens
- They can be produced from:
 - **Free-radical substitution** of alkanes
 - **Electrophilic addition** of alkenes
 - **Substitution** of an alcohol

Free-radical substitution of alkanes

- A free-radical substitution reaction involves **three main steps: initiation, propagation, and termination**
- This reaction occurs between an alkane and a halogen, such as **chlorine (Cl₂)** or **bromine (Br₂)**, in the presence of **ultraviolet (UV) light**
 - **Initiation:**
 - UV light provides energy to break the Cl-Cl or Br-Br bond by **homolytic fission**, producing two identical **halogen free radicals** (Cl• or Br•)
 - **Propagation:**
 - The halogen radicals react with alkane molecules in a **chain reaction**, producing new radicals and continuing the substitution of hydrogen atoms with halogen atoms
 - **Termination:**
 - The reaction stops when **two free radicals combine** to form a stable molecule, ending the chain process

Free-radical substitution reactions of alkanes



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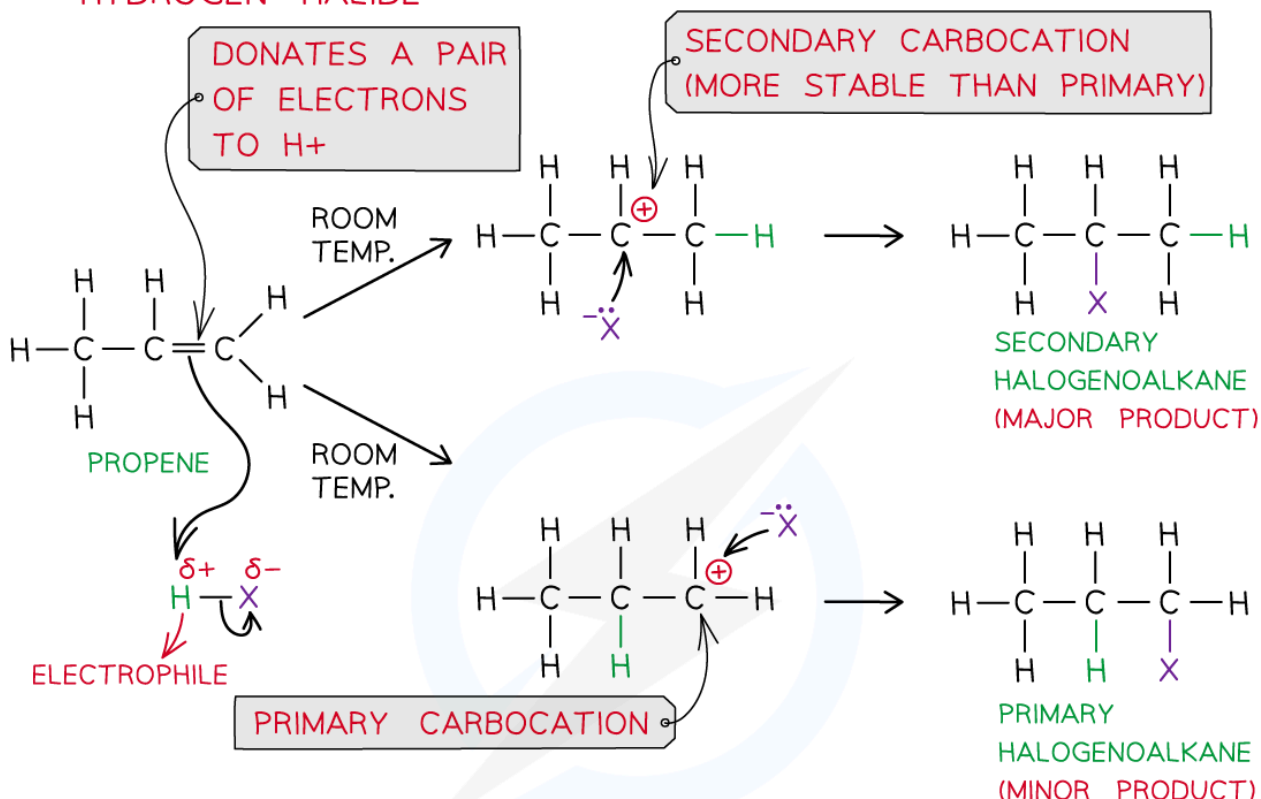
Free-radical substitution reactions of alkanes produce halogenoalkanes

Electrophilic addition

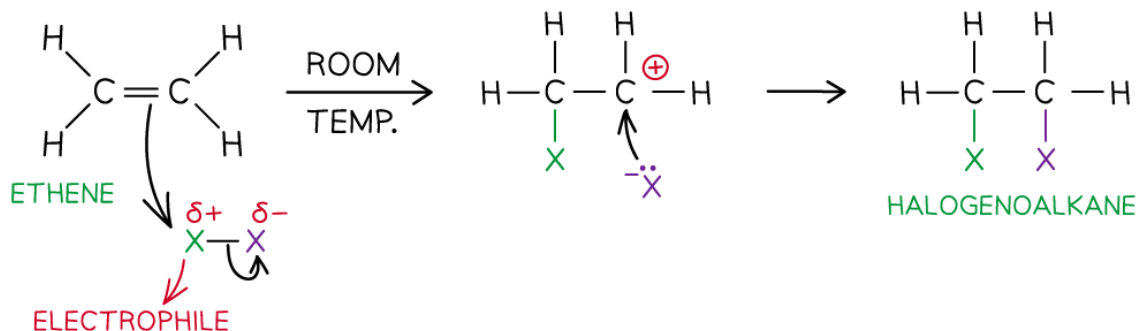
- Halogenoalkanes can also be produced from the addition of **hydrogen halides** (HX) or **halogens** (X₂) at **room temperature** to alkenes
- In hydrogen halides, the hydrogen acts as the electrophile and accepts a pair of electrons from the C-C bond in the alkene
 - The major product is the one in which the halide is bonded to the most substituted carbon atom (**Markovnikov's rule**)
- In the addition of halogens to alkenes, one of the halogen atoms acts as an electrophile and the other as a nucleophile

Electrophilic addition to alkenes

• HYDROGEN HALIDE



• HALOGEN



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Electrophilic addition of hydrogen halides or hydrogen at room temperatures to alkenes results in the formation of halogenoalkanes

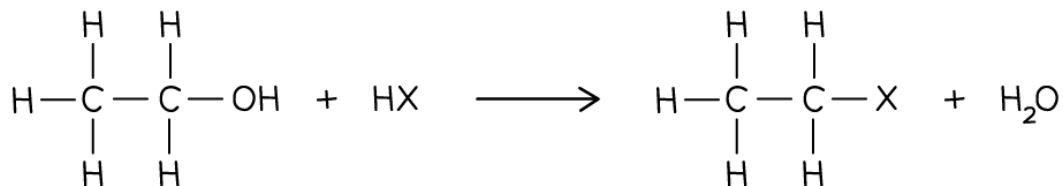
Substitution of alcohols

- In the **substitution of alcohols** an alcohol group is replaced by a halogen to form a halogenoalkane
- The substitution of the alcohol group for a halogen can be achieved by reacting the alcohol with:
 - HX** (or KBr with H₂SO₄ or H₃PO₄ to make HX)
 - PCl₃** and heat

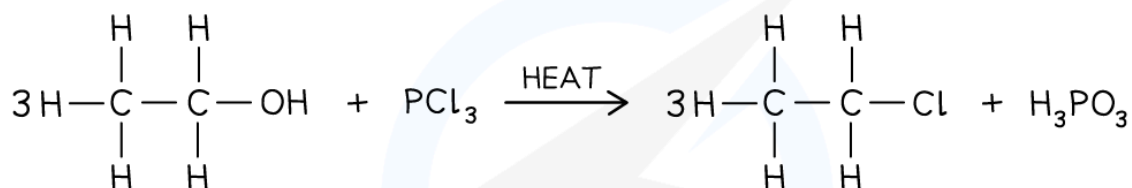
- PCl_5 at room temperature
- SOCl_2

Substitution of alcohols

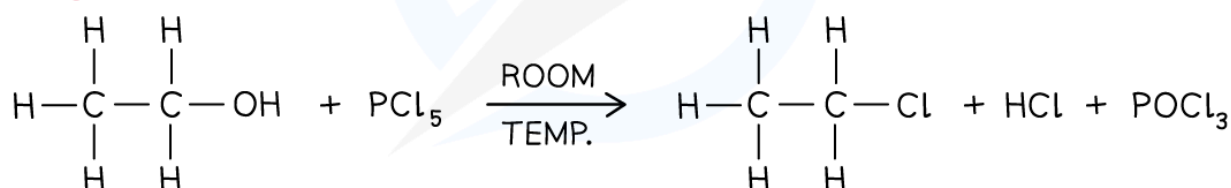
- HX (OR KBr WITH H_2SO_4 OR H_3PO_4)



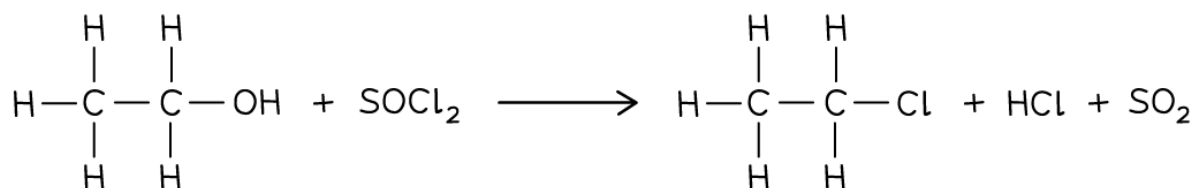
- PCl_3 + HEAT



- PCl_5



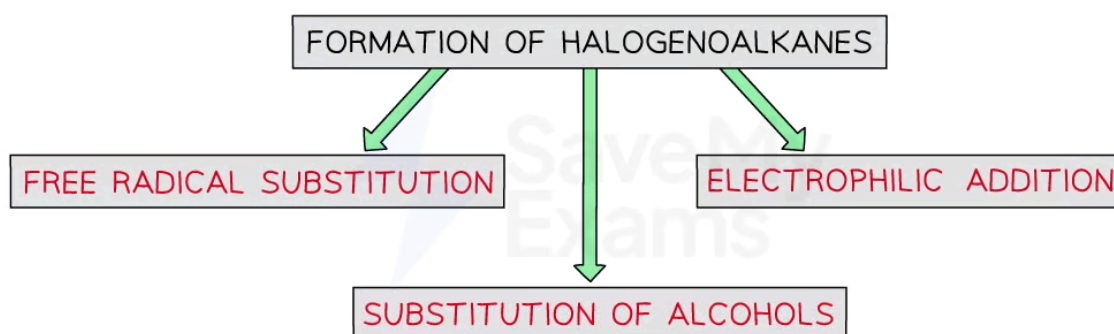
- SOCl_2



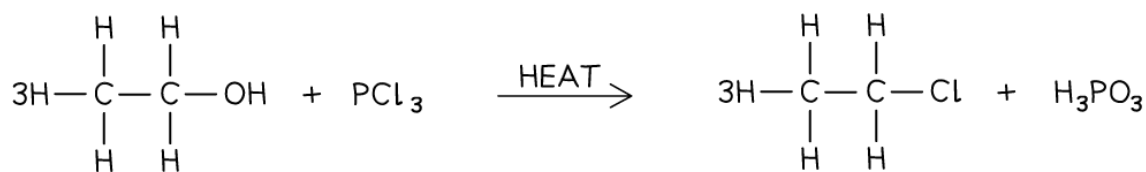
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Substitution of alcohols produces halogenoalkanes

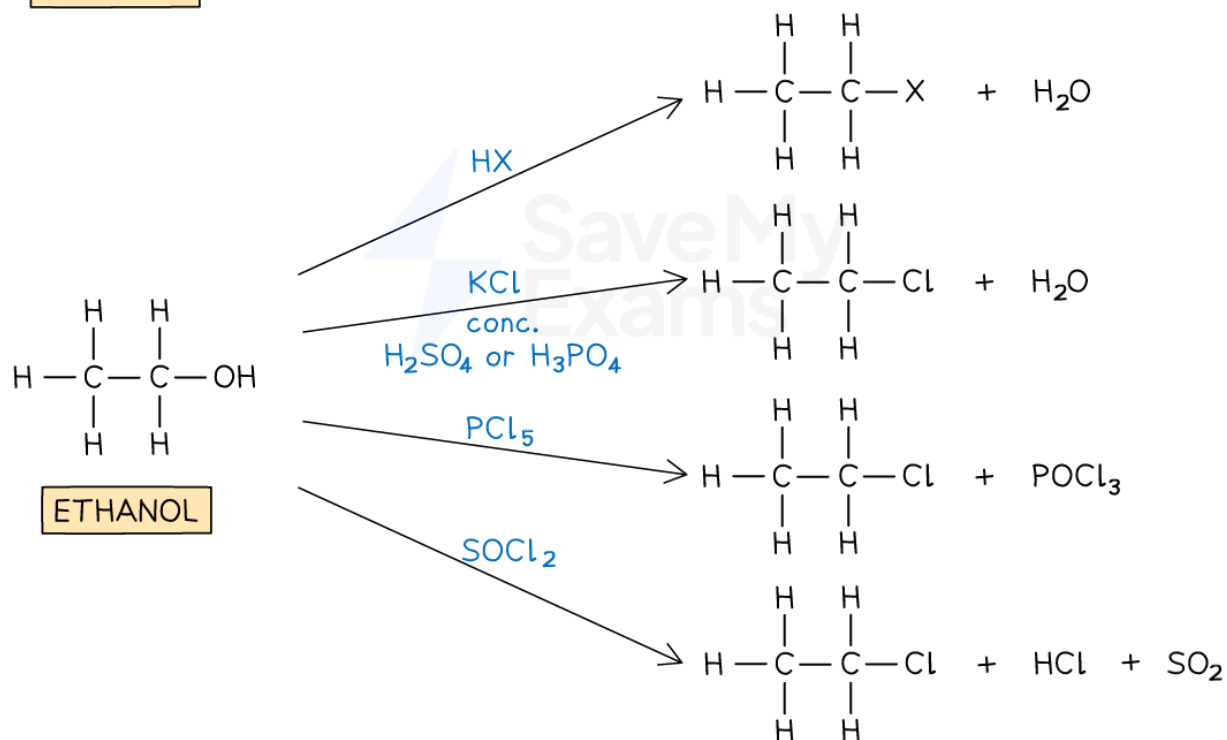
Different methods of forming halogenoalkanes



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ETHANOL



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Overview of the different ways to produce halogenoalkanes

Classifying Halogenoalkanes

- Depending on the carbon atom the halogen is attached to, halogenoalkanes can be classified as **primary**, **secondary** and **tertiary**
 - A **primary halogenoalkane** is when a halogen is attached to a carbon that itself is attached to one other alkyl group
 - A **secondary halogenoalkane** is when a halogen is attached to a carbon that itself is attached to two other alkyl groups
 - A **tertiary halogenoalkane** is when a halogen is attached to a carbon that itself is attached to three other alkyl groups

Primary, secondary and tertiary halogenoalkanes

The number of alkyl groups attached determines if the halogenoalkane is primary, secondary or tertiary